

Candidate Number _____

Fraction _____

2020 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

First Trial Session A

Zoological Experiments

An investigation and morphological record of the changes in behavior of brine shrimp in different environments.

Experimental equipment:

equipment	Quantity of equipment and	quantity
1. Dissecting microscope	reagents: 1 unit; 1. Normal brine shrimp	60ml
2. Tablet PC; 3.	solution 1 unit; 2. 15ml centrifuge tubes (equivalent to measuring cylinders)	2 units
5cm Petri dish (A); 4.	1. 3. 15ml of brine shrimp solution during starvation	
5cm Petri dish with black	1. 4. Mercury-treated brine shrimp solution, 15ml	
lid; 5. 2-partitioned 9cm Petri dish (B) (Square-hole black disc)	1. Label sticker (Exam number - Experiment A to D) -Number of repetitions in experiments 1-3)	1 sheet
6. 9 cm petri dish with black lid 7.	1	
One dropper (cut the large opening)		
8. One mercury dropper (cut the opening wide).		
9. One sheet of four-color cellophane (blue, red, yellow, and green).		
10. One sheet of black paper		
11. Transparent	1 sheet	
paper 12. One cup for mercury-treated brine shrimp waste liquid		
13. One non-mercury-treated brine shrimp waste liquid cup		
14. One pair of gloves		
15. Timer 16.	1	
Computer	1 unit	

Please note:

1. The materials and equipment on the table will not be replenished after they are used up.
2. This exam paper (including the cover and the exam paper) consists of 5 pages and must be returned in its entirety upon submission.
3. The total time allotted for this exam, including both Practice A and Practice B, is 90 minutes.
4. Please answer the questions on the answer sheet. You may write your answers on the back of the answer sheet, but please indicate the question number.

An investigation and morphological record of the changes in behavior of brine shrimp in different environments.

I. Introduction

Ecology is a discipline that studies the interaction between organisms and their environment. The environment broadly refers to the abiotic factors (temperature, humidity, pH, heavy metal pollutants, etc.) and biotic factors (predators, food, competitors, etc.) encountered by specific organisms. This experiment uses brine shrimp (*Artemia* sp.) as material to explore its behavioral changes in response to environmental factors.

In the marine environment, photosynthetic green algae are one of the main foods of brine shrimp. Brine shrimp are often used as fish feed in the market. Their ease of large-scale breeding and reproduction makes them frequently used in ecological and environmental research.

II. Experimental Instruments

1. Introduction to Dissecting Microscopes

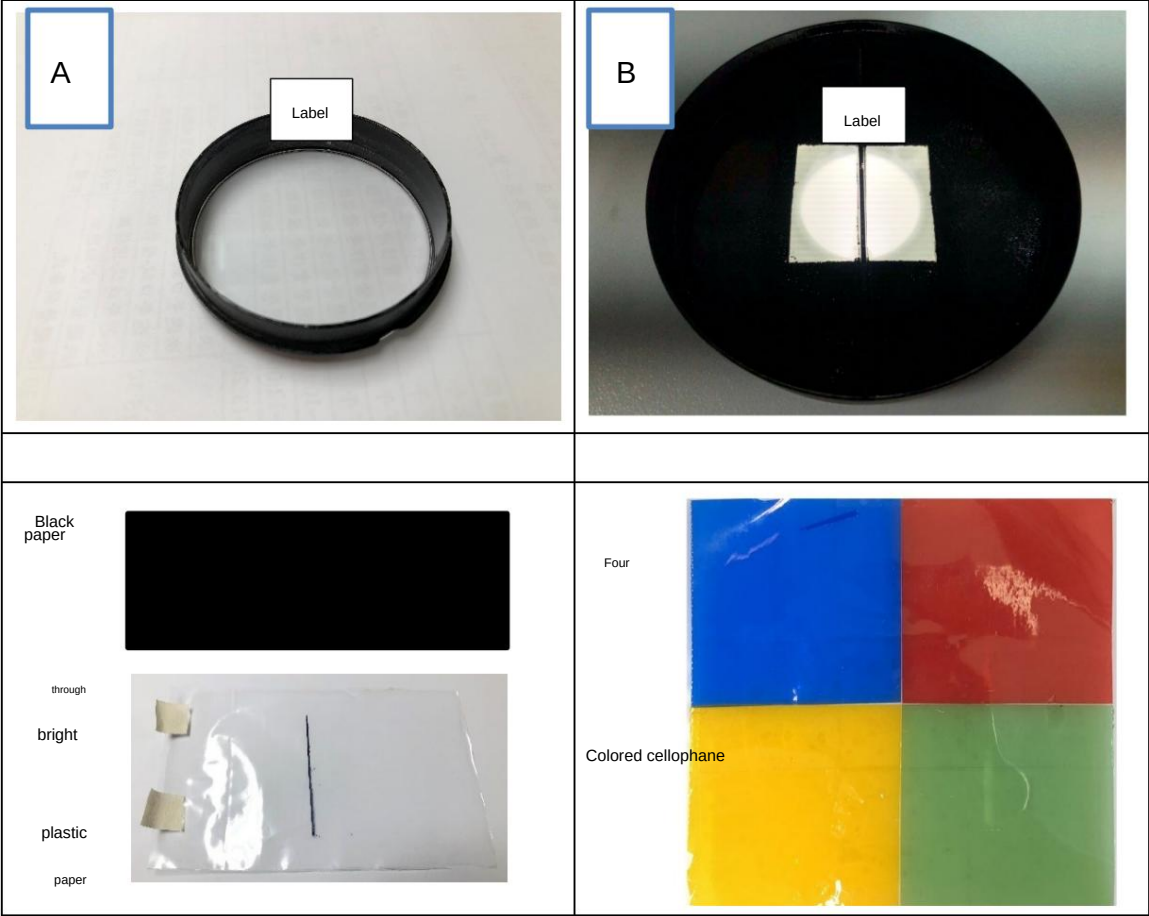


#Adjust the magnification wheel appropriately to observe the morphology of brine shrimp# Light source

adjustment knob: Adjust the light source to illuminate from bottom to top. Adjust the brightness to the highest setting for experiments A through D. #When observing brine shrimp in the petri

dish, please place the petri dish in the center of the light source to ensure that the calculated areas on both sides are equal when comparing experimental groups.

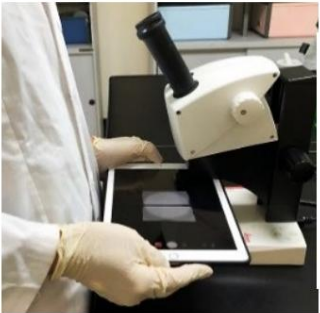
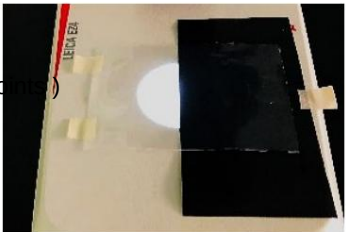
2. Petri dish numbering and paper strips



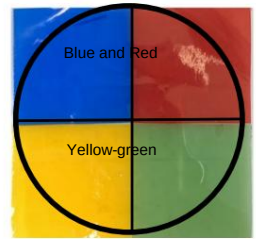
III. Experimental Items and Procedures:

Experiment A: Preference for light (whether or not one exhibits phototropism). (6 points)

1. Align the dividing line of the transparent plastic sheet with the center of the light source under the microscope and fix it in place.
 2. Place the black paper below the transparent plastic sheet, occupying half of the right side of the light source.
 3. Place petri dish A above the transparent plastic sheet, in the center of the light source, so that the light and dark areas are evenly divided.
 4. Use a dropper to add 5 ml of normal brine shrimp liquid to petri dish A and cover it with the black lid to allow the brine shrimp to adapt to the environment.
 5. After waiting 10 seconds, turn on the light source under the microscope. After irradiation for 30 seconds, open the lid, attach a label (exam number-A-number of repetitions), raise your hand and ask the teaching assistant to help you quickly remove the black paper, then tap the plate.
 6. Immediately press the shutter button (focus before the teaching assistant removes the plate).
 7. After taking the photo, gently shake the petri dish to distribute the brine shrimp evenly.
 8. Repeat the above steps, without repeatedly taking brine shrimp, until three sets of data (three replicates) are collected. Please remember to relabel and photograph each experiment, changing the label according to the number of replicates on the label. Finally, pour the waste liquid from the non-mercury-treated brine shrimp into the waste cup.
 9. The experiment is now complete. Please proceed to the next experiment.
- Use the tablet software "Doodle" to count the number of brine shrimp in the dark/bright areas (the circular area covered by the light source) in the photos and fill in the table on the answer sheet.



- Note the method of taking photos:
- The lens is placed
- Top of the petri dish
- Parallel to the petri dish
- Do not tilt



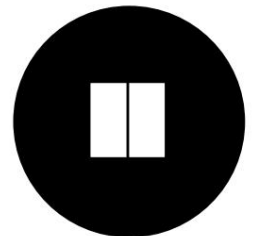
Experiment B: The preference of brine shrimp for different wavelengths of light. (6 points)

1. Place four colored cellophane sheets in the center of the light source, ensuring that the light source area occupied by each of the four zones is equal. 2. Place culture dish A on top of the colored cellophane sheets, ensuring that the light source area occupied by each of the four zones is equal. 3. Use a dropper to add 5 ml of normal brine shrimp liquid to culture dish A and cover it with the black lid to allow the brine shrimp to acclimate to the environment. 4. After waiting 10 seconds, turn on the light source below the microscope. After irradiating for 50 seconds, open the lid and affix a label (examination number-).
5. After taking the photos, gently shake the petri dish to distribute the brine shrimp evenly. 6. Repeat the above steps, without repeatedly taking brine shrimp, until three sets of data (three replicates) are collected. Please remember to relabel and photograph each experiment, changing the label according to the number of replicates on the label. Finally, pour the waste liquid from the non-mercury-treated brine shrimp into a container. 7. Use the tablet software "Doodle" to count the number of brine shrimp in each of the blue/red/green/yellow regions in the photos, and fill in the data.

Enter the information into the table on the answer

sheet. 8. The experiment is now complete. Please proceed to the next experiment.

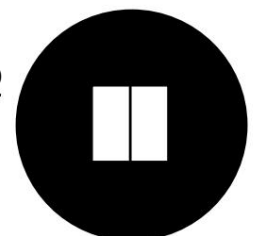
Starving group Normal group



Experiment C: The degree to which brine shrimp prefer light when they are hungry. (6 points)

1. Place culture dish B in the center of the light source, ensuring equal illumination on both sides. 2. Using a dropper, add 5 ml of the starved shrimp liquid and 5 ml of the normal brine shrimp liquid to each side of culture dish B.
- The left side represents the starved group, and the right side represents the control group with normal brine shrimp. A black lid was placed over the brine shrimp to allow them to acclimatize. 3. After waiting 10 seconds, turn on the light source below the microscope. After 30 seconds of illumination, open the lid and attach a label (examination number-C-number of experimental replicates). Take a photograph with a plate. 4. After taking the photograph, gently shake the petri dish to distribute the brine shrimp evenly. 5. Repeat the above steps, without repeatedly taking brine shrimp, until three sets of data (three replicates) are collected. Please remember to relabel and photograph the dish after each experiment, changing the label according to the number of experimental replicates. Finally, pour the waste liquid from the non-mercury-treated brine shrimp into a non-mercury-treated brine shrimp waste cup.
5. Using tablet software to "doodle" and statistically analyze the presence of light on both the hungry and normal sides of the brine shrimp in the photos.
6. The experiment is now complete. Please proceed to the next experiment.

Mercury treatment normal group



Experiment D: Does the presence of mercury ions (a heavy metal) affect the light preference of brine shrimp? (10 points)

1. Please ensure you are wearing gloves. 2. Place culture dish B in the center of the light source, ensuring the illumination area on both sides is consistent. 3. Add 5 ml each of the mercury-treated solution (use a mercury-specific dropper) and the normal brine shrimp liquid to culture dish B.
- The left side shows mercury-treated brine shrimp, while the right side shows normal brine shrimp as the control group. Both groups are covered with black lids to allow the brine shrimp to adapt to the environment.
4. After waiting 10 seconds, turn on the light source below the microscope. After irradiating for 30 seconds, open the lid and attach the label (examination number-D-number of experiment repetitions). Take a photo with a flat plate.

5. After taking the photos, gently shake the petri dish to distribute the brine shrimp evenly. 6.

Repeat the above steps, without repeatedly taking brine shrimp, until three data collections (three replicates) are completed. Please remember to relabel and photograph the sample after each experiment, changing the label according to the number of replicates indicated on it. Finally, pour the waste liquid from the mercury-treated brine shrimp into the container. (5 points will be deducted for improper disposal and failure to recycle according to regulations).

7. Using tablet software to "doodle" and statistically analyze the presence of light on the mercury-containing and normal sides of brine shrimp in the photos.

Please indicate the number of items distributed in each area below and fill in the table on the answer sheet.

Discussion 1: Please conduct experiments A through D in sequence, and draw conclusions regarding the relationship between brine shrimp and the environment for each experiment. (12

point)

Discussion 2: Based on the above experiments, propose a hypothesis regarding the food chain relationship between brine shrimp and predators in the environment. (10 **points**)

Please fill in the above discussion on the answer sheet.

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First Trial Session B

Zoological Experiments

Observation of earthworm structure

Equipment 1 (Please leave it on the table when changing locations)		Equipment 2 (One set per person, no reuse)	
name	Quantity	Name	quantity
Dissecting microscope,	1 unit,	Rubber gloves,	1 pair,
pointed forceps,	2 sticks,	insect pins (in a flashing counter bottle), eight	8
pointed scissors,	1	colored pins (inserted into a tech sponge), tech sponge (for use	sticks,
tissue paper	bundle, 1 pack	as a dissecting tray), earthworms (in a 15	10
		mL centrifuge tube), water (in a 15 mL centrifuge	sticks,
		tube), dropper (3 mL).	1
			piece, 1 tube, 2 sticks

Please note: 1.

Materials and equipment on the table will not be replenished after use. 2.

This exam paper (including cover and question paper) consists of 11 pages and must be returned

upon submission. 3. The total time for this exam, including Practical A and Practical B, is 90

minutes. 4. Please answer on the answer sheet. Answers may be written on the back of the answer sheet, but please indicate the question number.

Earthworms are a general term for large terrestrial species belonging to the class Annelidae, subclass Oligochaeta, phylum Annelida. Approximately 6,000 species are currently known.

There are many species, with over a hundred known in Taiwan. Earthworms' bodies are composed of many segments. In mature earthworms, the epidermis of certain segments thickens and swells, and may even fuse to the point where the individual segments are indistinguishable. This structure is called the clitellum, which plays a crucial reproductive role in producing egg cocoons. The location and morphology of the clitellum are important characteristics for identifying earthworm groups.

One of the signs.

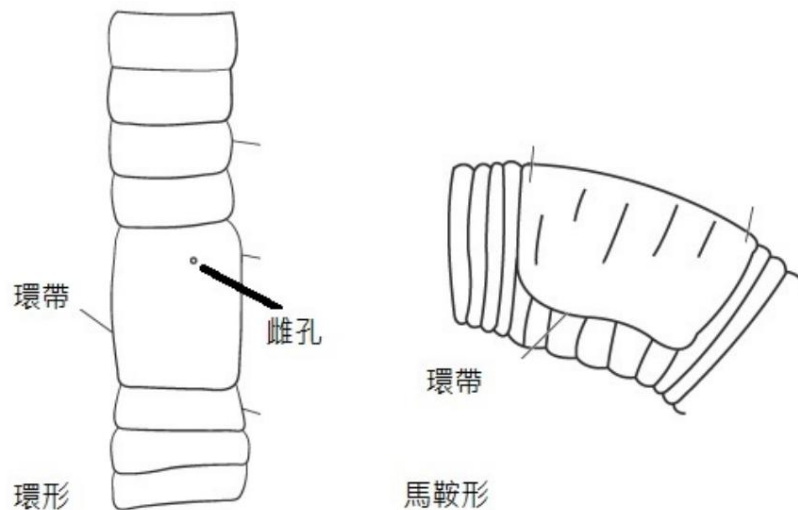
To the average person, earthworms appear similar in appearance and internal structure. However, earthworms from different groups can still have significant differences in many characteristics. Therefore, diagrams showing the external morphology and internal structure of earthworms can only be used as a reference and cannot guarantee that the location, number, size, appearance, and structure of the morphology and structure will be completely consistent.

Please carefully observe and answer the related questions using the equipment and earthworms provided on the table.

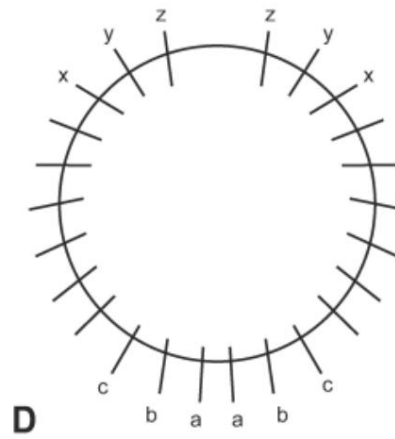
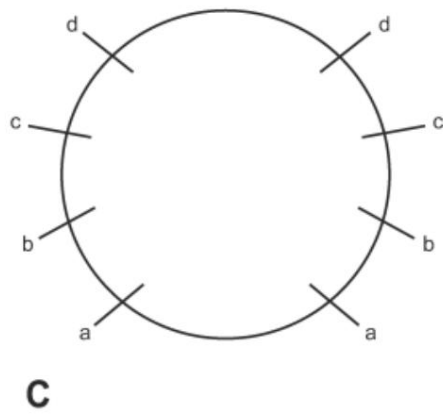
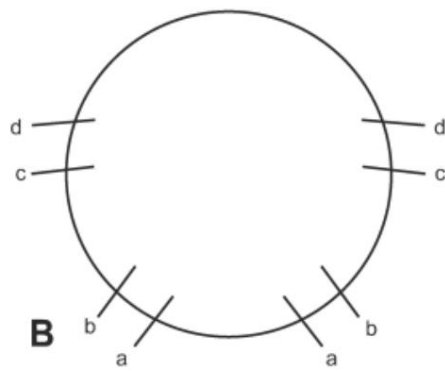
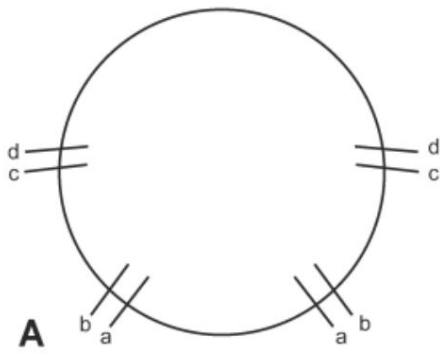
The position of the **circumference** ranges from segment 1 to segment 2 (1 point each).

Please circle the correct circumflex morphology: circumflex/saddle-shaped (1 point), non-segmented fusion/segmented non-fusion (1 point)

point)

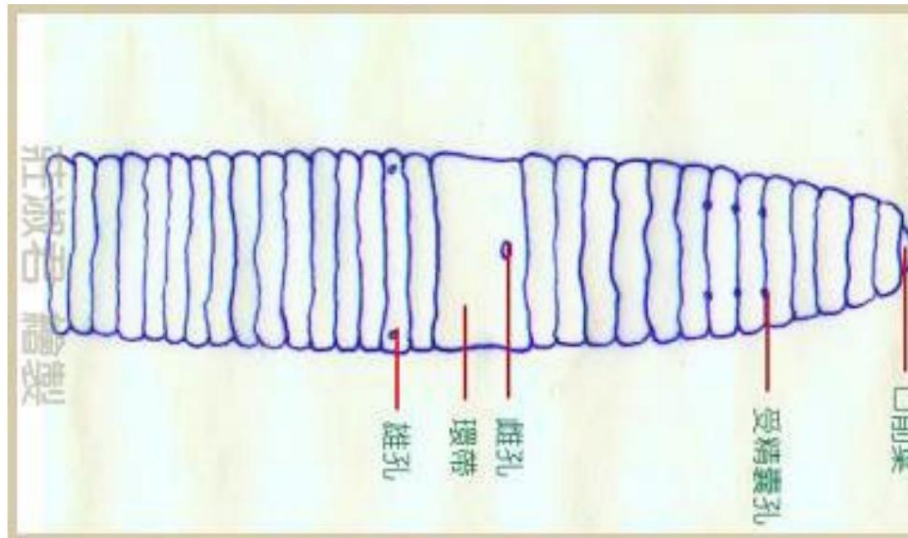


The setae morphology of this earthworm belongs to which of the following types: _____ (1 point)



Earthworms are hermaphroditic animals that reproduce through cross-fertilization. Starting from the anterior end of the body and proceeding down the ventral side, the spermatheca pore and the female reproductive organ are visible in sequence.

Earthworms have three groups of reproductive pores: male pores, female pores, and male pores. Although the number, location, and morphology of each pore vary among different species and types of earthworms, their arrangement remains constant. During mating, both male pores release sperm into each other, while the seminal vesicles receive the sperm. Therefore, earthworms adopt a reverse parallel posture during mating, with the male pores and seminal vesicle pores facing each other to facilitate sperm transfer. The female pores are used for egg production during reproduction and are not used during mating.

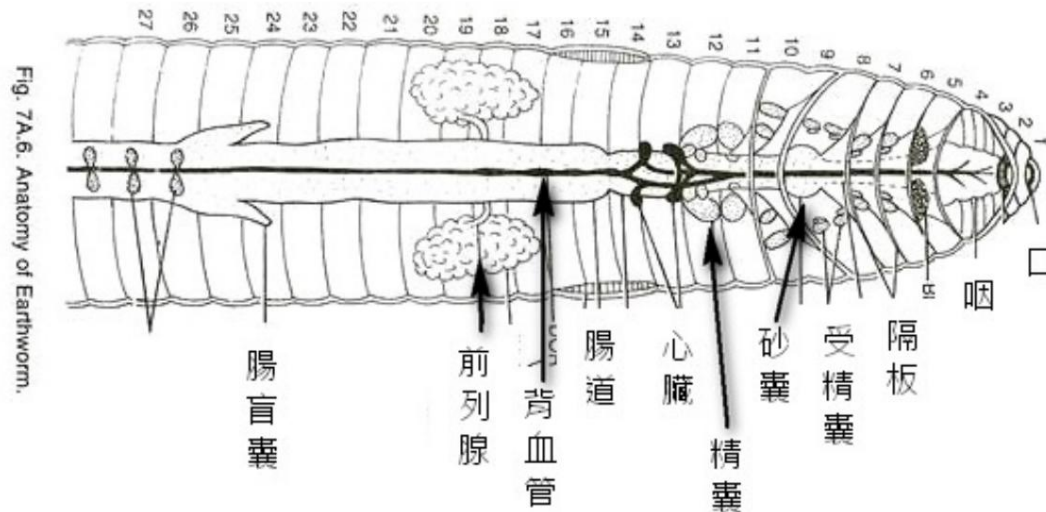


Please carefully refer to the diagram of the earthworm's genital opening and observe a close-up photo of this type of earthworm mating, and answer the following questions.

The foramen of the spermathecal receptacle is located in segment ____ (4 points)

The female opening is located in the ____ segment (4 points).

The male aperture is located in section ____ (2 points).

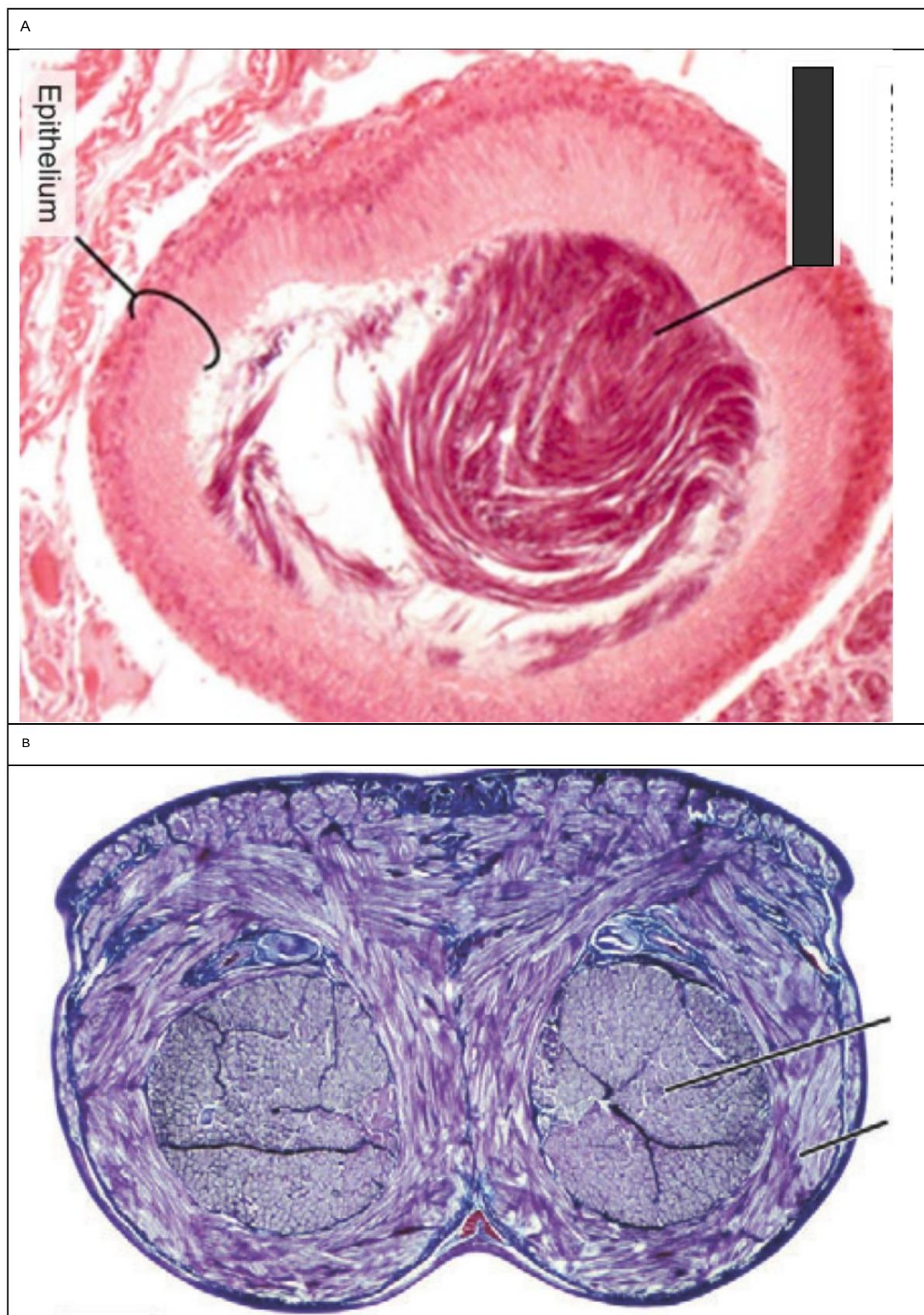


Please operate under a dissecting microscope, referring to the diagram above showing the internal structure of an earthworm viewed from its dorsal side, and locate this type of earthworm.

Mark the locations of the following organs with colored pins, and finally cut the marked earthworm organs with the tissues listed below.

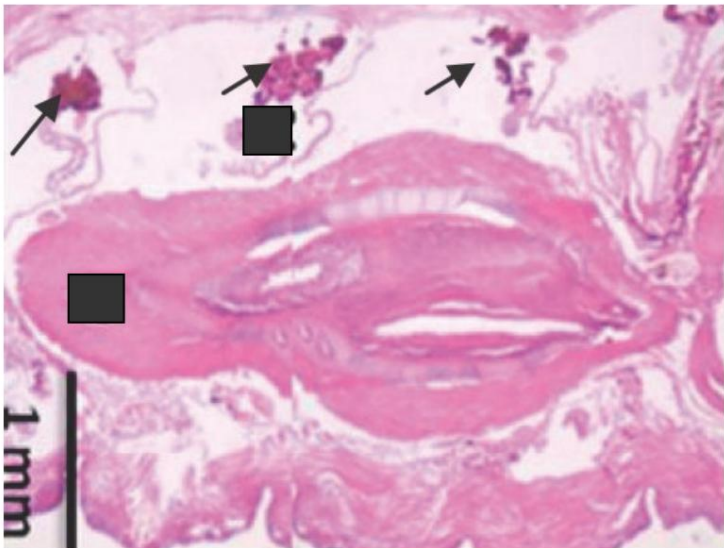
The photos should correspond to each other. Please pay special attention to the requirements for the remarks; if the remarks are not completed, only half the marks will be awarded for this section.

Organ or structure	pin color	tissue	slide photograph code	remarks
Gizzard	Red (1 point)		(1 point)	
Seminal vesicles	Yellow (2 points)		(3 points)	
Prostate	Green (1 point)		(3 points)	
Blue spermatophore (3 points)			(3 points)	
Dorsal vein	Peach (2 point)		(3 points)	Mark sections 30-35 with two pins.
Ventral nerve cord	White (2 points)		(3 points)	Mark any 5 sections with two pins.
Intestinal wall	Purple (1 minute)		(3 points)	Please note that the pin must be inserted in a cutting motion. Viewing perspective
Silver in the circumferential body wall (1 point)			(3 points)	Please note that the pin must be inserted in a cutting motion. Viewing perspective

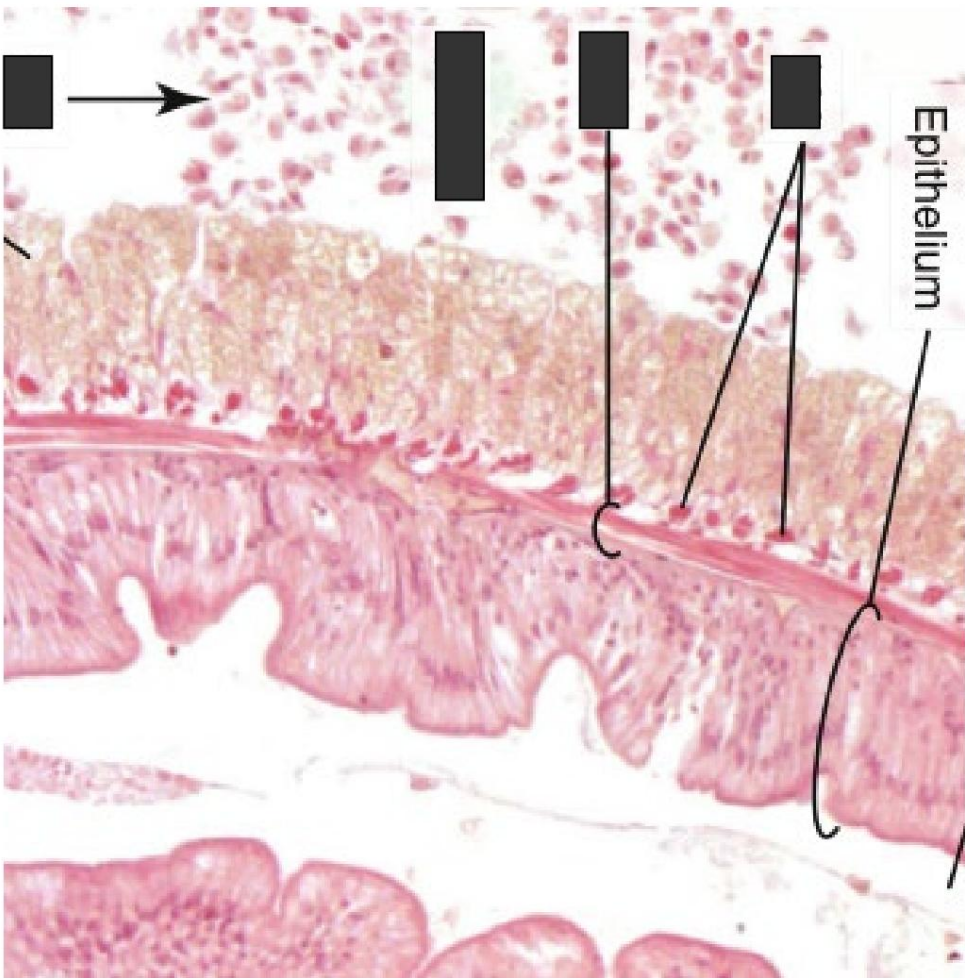


Epithelium: epithelium

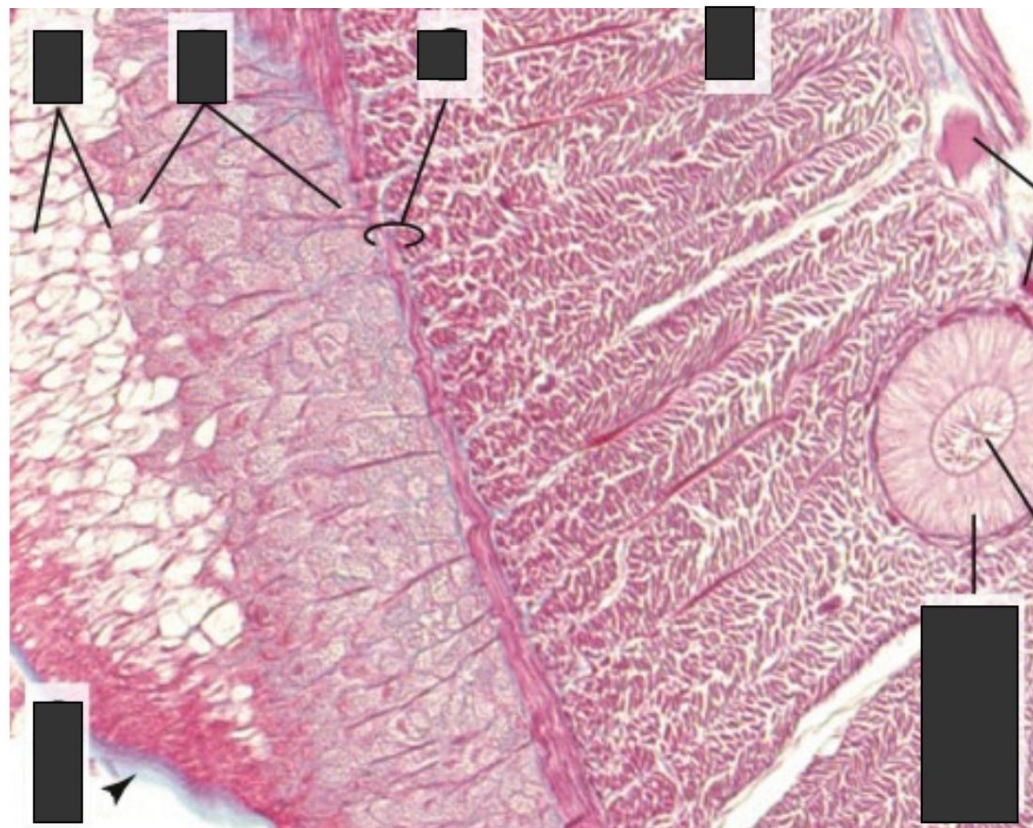
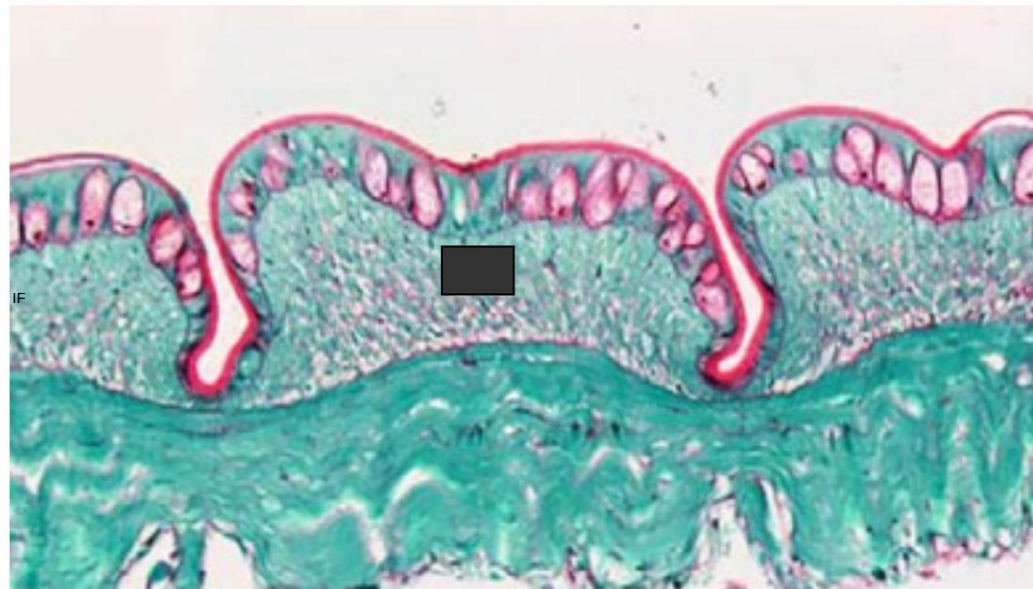
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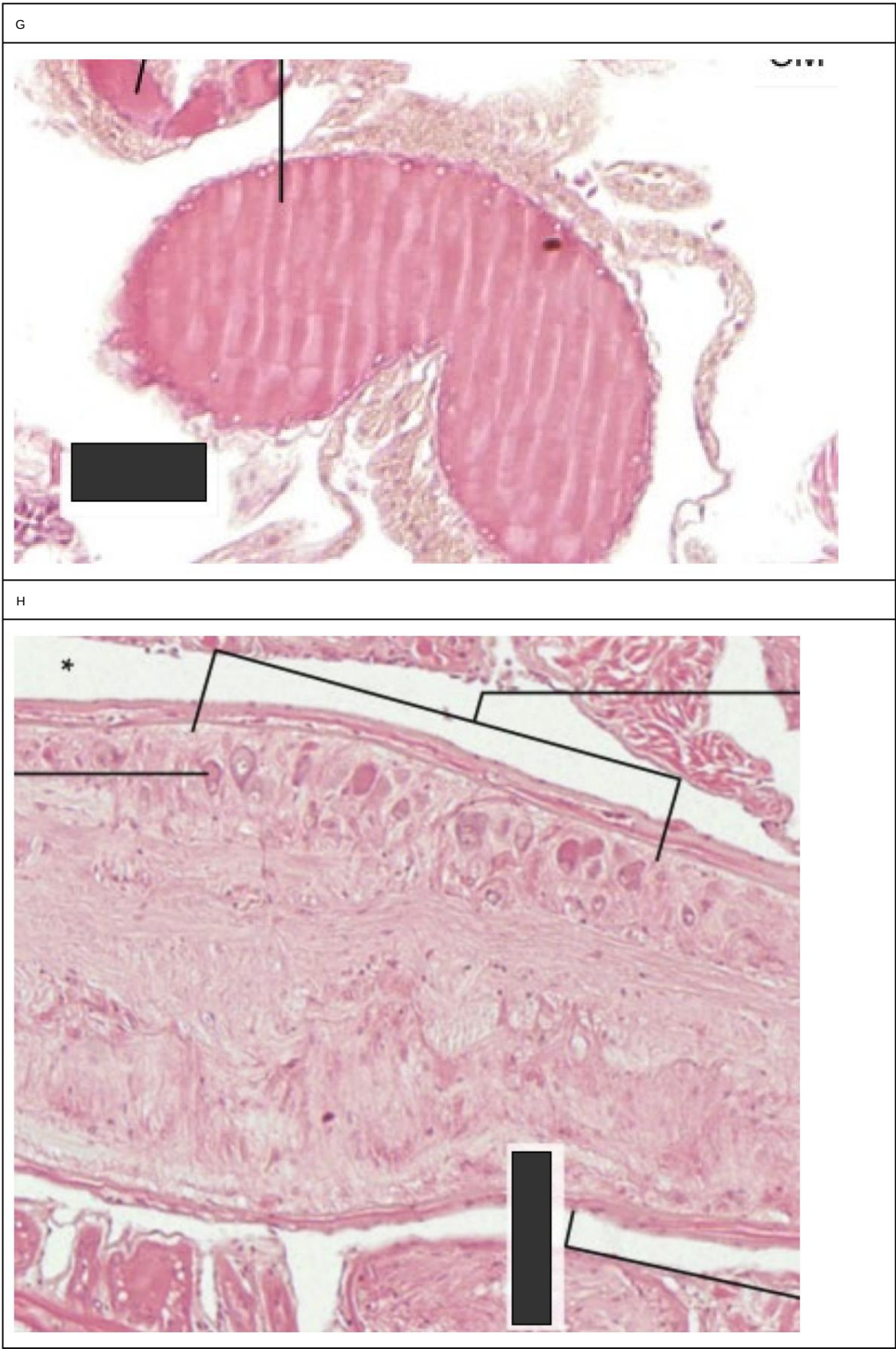


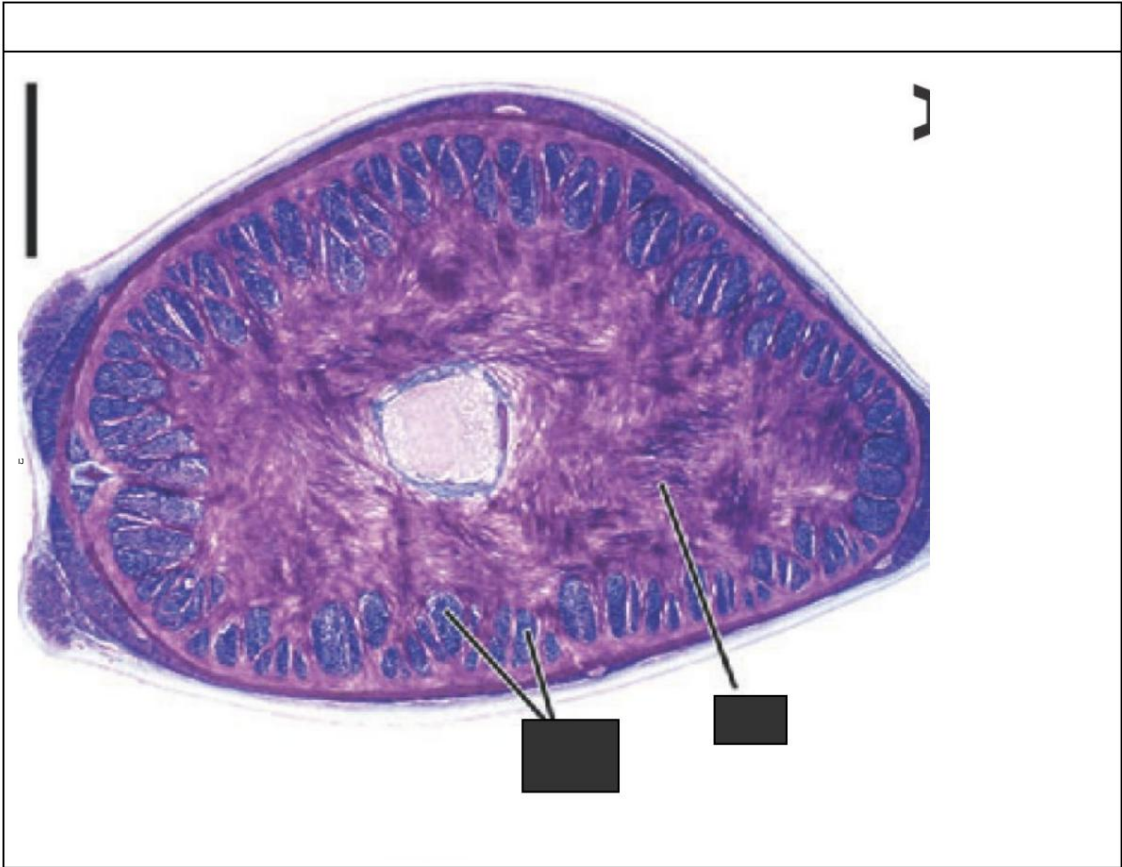
d



Epithelium: epithelium







K



L

